

# BHRUGU SETLUR

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## EDUCATION

### New York University – Tandon School of Engineering

New York, NY

*B.S. Electrical and Computer Engineering | GPA: 3.74 | Dean's List: all semesters, 2025–2026*

*Expected May 2028*

- Coursework: Data Structures & Algorithms (Python), Object-Oriented Programming (C++), Computer Architecture & Digital Logic (Verilog)

## EXPERIENCE

### Tokonoma.ai

San Francisco, CA

*Software Engineering Intern*

*Jun 2026 – Present*

- Develop an AI SRE automation platform: an MCP (Model Context Protocol) server giving LLM agents natural-language access to production logs, distributed traces, and Kubernetes events.
- Designed the vector-similarity deduplication system for the agents' knowledge base — HNSW-indexed cosine search over pgvector embeddings flags near-duplicate incident records at save time — adopted for the production implementation.
- Cut agent incident-investigation time 53% (15 to 7 turns on evals) by re-ranking semantic search to surface proven runbooks ahead of raw incident history.
- Expanded the agent eval suite to 54 fault-injection scenarios and enriched trace search from partial spans to full call trees; closed prompt-injection, file-access, and OAuth 2.1 vulnerabilities.

Tech: Python, Rust, PostgreSQL, pgvector, Kubernetes, Helm, Docker, OpenTelemetry, Quickwit, OAuth 2.1, MCP, RAG

### Generex

San Francisco, CA

*Embedded Firmware Engineer (Contract)*

*Jun 2026 – Present*

- Developing replacement firmware for an industrial battery-monitoring sensor: an STM32 microcontroller (ARM Cortex-M0+) reads 4 laser time-of-flight sensors to detect low fluid levels and reports alarms over RS-485.
- Wrote the protocol and alarm logic as portable, hardware-independent C, making it fully testable off-device; 354 automated checks pass before the board is even powered on.
- Designed alarm logic that resists false alarms: readings are debounced with hysteresis, and a time-based confidence model distinguishes a genuinely low battery cell from condensation, bubbles, or a detached sensor.

### Growcery – growcery.us

New York, NY

*Software Engineering Intern*

*Jan 2026 – May 2026*

- Built and launched a full-stack marketplace with buyer-seller accounts, in-app messaging, and seller listing tools.
- Integrated Stripe marketplace payments with Apple Pay support and automated 10% platform fee collection.
- Implemented radius-based geolocation discovery with Supabase and Mapbox; integrated Spoonacular API; deployed on Vercel.

Tech: Next.js (React, TypeScript), Node.js, Supabase, Stripe, Mapbox API, Spoonacular API, Resend API, Vercel

### VIP – High Performance Computing at NYU

New York, NY

*Virtual Environment Penetration Tester*

*Sep 2025 – Present*

- Executed an 8-phase attack kill chain against Google Cloud research environments (SRDEs) — exploiting SQL injection, command injection, local file inclusion, sudo privilege escalation, and credential reuse to gain root.
- Automated Python log-anomaly detection and MD5 hash cracking; presented severity-ranked remediation plans.

## PROJECTS

### 8-bit CPU with Virtual Memory & Cache – Verilog | github.com/bhrugusetlur-art/cpu-design

Jun 2026 – Present

- Designed the RTL for a complete 8-bit Harvard-architecture CPU — 12-instruction ISA, ALU, register file, control unit, datapath — with a write-back L1 direct-mapped + L2 4-way set-associative (LRU) cache hierarchy.
- Built a virtual memory system — a 4-entry TLB and cache-mediated MMU with hardware page-table walks and page-fault handling; verified all 13 modules in simulation and on a Basys 3 (Artix-7) FPGA, targeting physical tapeout next.

Tech: Verilog, RTL Design, FPGA (Xilinx Artix-7), Vivado, Icarus Verilog, GTKWave, C++

### LineLab – NYC Transit Planning | github.com/Tandonites/LineLab

Apr 2026

- Built a web app where users draw NYC subway corridors and simulate ridership, operating cost, and travel time.
- Trained XGBoost models on 4 years of MTA ridership + Census/OSM data; built a NetworkX GTFS graph for shortest-path travel time, integrating NY Open Data and US Census APIs for ML inference at map-draw time.

Tech: React 19, TypeScript, Vite, Leaflet, Python, FastAPI, NetworkX, XGBoost, MTA GTFS, NY Open Data, US Census API, OSM

## SKILLS

**Languages:** Python, C, C++, Rust, TypeScript, JavaScript, Java, C#, SQL

**AI/ML:** RAG, semantic search, vector embeddings (pgvector), LLM agents, Model Context Protocol (MCP), prompt engineering

**Frameworks & Tools:** React, Next.js, Node.js, FastAPI, PostgreSQL, Supabase, Kubernetes, Helm, Docker, OpenTelemetry, Git, GCP, Nmap, sqlmap

**Hardware & Embedded:** Verilog, RTL Design, FPGA (Xilinx Artix-7), Computer Architecture, STM32 (ARM Cortex-M0+), I2C, UART/RS-485, Arduino, Circuit Design, Soldering

## AWARDS & HONORS

AP Scholar | Congressional Gold Certificate | **3rd Place – Code4City Hackathon** | 6th Place, USA Fencing Div. II Nationals